

1.0 INTRODUCTION

In order to implement the White House's 18 December 2025 Executive Order, "Ensuring American Space Superiority" in a timely and cost-effective manner, NASA is undertaking a new pathway to implement a free flying space station that establishes a continued U.S. crewed presence in LEO. This future space station is envisioned to start as a publicly owned and operated U.S. Space Station that ensures a platform with frequent crewed access, advances human space exploration, scientific discoveries, and serves as a springboard for commercial expansion and growth to maintain United States economic competitiveness.

The strategy includes an extensible space station that provides commercial providers the ability to provide research and other commercial services as soon as they come online by commercial activities on the U.S. Space Station through their own commercial modules, private astronaut missions, or smaller business development efforts utilizing interior or exterior resources made available by the U.S. station. As commercial markets mature, NASA's goal is to support an evolution to commercially owned and operated space station(s) and be one of many customers.

Maintaining a crewed presence in LEO and avoiding a transportation gap between International Space Station (ISS) and a future station is challenging with the nature of new development and the limited funding to develop a new station while ISS is operational. Crew and cargo transportation gaps of greater than a year can result in loss of the company's ability to support and fund critical infrastructure and to maintain skillsets, elevating the risk of the company folding or walking away from the capabilities. As NASA has learned, these capabilities are not easily restarted. An added complexity is the transportation costs for crew and cargo which are the majority of the costs for ISS and will be for a future station as well. Overlapping ISS and a future station close to doubles the transportation demands which becomes very costly.

To maintain on-orbit US crewed presence, eliminate significant transportation gaps or overlaps, and make maximum use of the United States investment in the ISS, the new space station should leverage the ISS and its related on-orbit resources, hardware, technical skillsets and transportation capabilities. The foundation of this new space station will be a NASA core module, that will be capable of attaching to the ISS or operating independently at 51.6 degrees inclination. While at ISS, this core will leverage ISS Crew and Cargo Vehicles for initial activation and outfitting including hardware transfer from ISS and provide early commercial opportunities. Opportunities will be offered while the core is at ISS for commercial activities, including the inclusion of two commercial modules. The evolved approach after ISS separation will offer multiple opportunities for industry growth of additional commercial modules as markets developed.

NASA will continue to procure Commercial Space Transportation services (Crew and Cargo). The plan will ensure transportation capabilities remain viable between the arrival of the new core station attached to ISS and the end of the ISS. Using ISS as a staging platform allows for shared use of these transportation assets to meet both ISS and the future station's initial demands. It also allows for incremental assembly and outfitting and on-orbit checkout leveraging existing investment in ISS systems and transportation capabilities.

- Leverage ISS crew missions and cargo flights to ISS to support outfitting and assembly

- Allow for less complex initial system capability
- Allows for more operational flexibility in problem solving, contingency planning and use of ISS as a safe haven
- Ensures smoother transition between platforms

This concept of operations (Con Ops) summarizes NASA's plans to provide commercial access, growth and expansion capability using the ISS as the initial location for a new LEO platform that will begin with an initial "Core Module", ability to add two commercial modules, followed by separation into a free flying LEO space station. The concept of operations assumes an evolutionary architecture that will flex to meet the needs of NASA and commercial partners. This document captures the potential mission architecture and high-level descriptions of the roadmap needed to establish this capability.

The document also includes incremental configurations and capabilities for the potential mission architectures, the concept for its assembly, the major architectural elements and interfaces, commercial opportunities, the utilization and expected operations during each incremental phase and the expanded commercial additions that will be considered as part of the development leading to the final configuration.

1.1 FUTURE LEO OBJECTIVES

NASA intends to maintain American space superiority by using the ISS to directly evolve to the next continuously crewed, government owned space station that meets NASA's operational, research and testing needs, and serves as a platform for commercial expansion. This platform is intended to be a continued catalyst for commercial industry and the growth of a LEO economy, to collaborate internationally and to conduct critical research for exploration and benefits on earth. NASA's objectives in LEO are:

- **Ensure American Space Superiority** – Continuous crew with operational, research and test capabilities
 - Moon to Mars Critical Technology - Test and evolve micro-g sensitive technology to reach performance and reliability targets
 - Human Health Research - Develop mitigations for long duration spaceflight-driven human health challenges
 - Impacting Research – Conduct NASA, commercial or other government agency funded research with high impact potential
- **Foster Economic Growth & Commercial Pathways** - In accordance with EO on maintaining space superiority
 - Commercial Expansion & Demand Stimulation – Open multiple opportunities for commercial access, addition of commercial modules or segments and an evolution path to an independent commercial station
 - Pivotal Manufacturing Applications - Develop in-space or space-derived manufacturing capabilities for cutting edge applications (e.g. silicon wafers, optical fibers, human retina manufacturing, etc.)
 - Break-through Technologies - Advance technologies such as major disease treatment (e.g. cancer, motor-neuron, etc.) and tissue/organ printing

- **Enable International Collaboration** – Collaboration or contribution opportunities that benefit the US, continuing US leadership in LEO

NASA's plan in LEO will prioritize American superiority in space with NASA providing base capabilities to support the maturation of a LEO market:

- NASA will make available a government Core LEO platform which includes crew and cargo transportation
- Stimulate growth opportunities for US Commercial providers for both the demand and supply sides. Offer opportunities for commercial participation on the Core LEO platform while at ISS and during free flight in LEO including the inclusion of two commercial modules.
- Invest in cutting-edge technology that will help spur the LEO economy as well as provide an investment for moon to mars
- Enable LEO market demand by making ISS more commercially accessible with more research and flight opportunities
- Continue to offer flight opportunities for Private Astronaut Missions

1.2 HIGH LEVEL ARCHITECTURE CONSIDERATIONS:

This architecture leverages the ISS for initial buildup and checkout of the initial elements, then separation prior to ISS end of life to become a free flying space station that can support one or multiple commercial modules/elements. Key drivers for the architecture of the future space station include the capability for 4 continuous crew with capability for direct crew handovers, NASA's future objectives in LEO, extensibility for expansion for commercial modules both at ISS and after separation, the inclusion of robotic and/or EVA capability, ISS architecture-driven compatibility and launch vehicle capabilities.

The concept centers around an initial Core Module that will be launched as the first new element, that can be evolved later. The core module will include propulsion, propulsion resupply, initial power capabilities, communications, command and data handling, navigation, avionics/software, basic crew habitation and heat rejection. This core module would launch and rendezvous with the ISS using onboard capabilities and then perform proximity operations to allow ISS capture using the SSRMS. The core module's forward port would contain a Common Berthing Mechanism to allow for connection to the Node 2 Forward port on ISS. The ISS currently has a docking port at the Node 2 Forward port, requiring reconfiguration to move the docking port to another ISS location, leaving the ISS with a berthing port at Node 2 Forward.

After installation of the Core Module, two commercial modules can be attached while at ISS. These modules would have research and other habitable and stowage spaces, with the ability to store ISS transferred hardware. If less than two commercial modules were available, then the existing ISS Permanent Multi-logistics Module (PMM) could be used in lieu of one or both to provide large equipment transfer from ISS and serve as additional stowage module for the new station. While at ISS, ISS power, thermal and other resources would be available for the core and commercial modules, within limitations of the existing systems. Commercial modules would bring additional power generation and heat rejection capabilities to ensure full use of these modules though available space while attached to ISS limit the allowable configurations. These commercial modules could serve in a variety of different

commercial arrangements to allow for internal and external opportunities to their capability. Feedback from commercial partners will help shape what basic services and configuration considerations need to be factored into the design.

After separation from the ISS, the new Core Space Station will be a self-sufficient free flying crew tended platform that can evolve into a permanently crewed space station or serve to host long duration sorties, housing visiting vehicles and crews as the demand and market matures. A second module called the Power and Cooling Module, with solar arrays provides additional power, stowage, and heat rejection would be added after the station becomes a free flyer to add sufficient power and heat rejection capabilities to support complex regenerative life support systems for continuous crew presence and to reduce the transportation vehicle demand.

Once separated from ISS, given ISS lessons learned for vehicle traffic and resupply needs to support 4 continuous crew and the ability for direct crew handovers (surge to 8 crew for short durations), the architecture will need to maintain sufficient docking/berthing ports dedicated to visiting vehicles. Depending on port traffic for NASA needs, these ports could be available for use by commercial providers in addition to NASA needs.

The current launch vehicle market offers opportunities over the Space Shuttle due to the advent of significant heavy launch capabilities and larger fairing sizes. This allows for new volumetric solutions previously unavailable due to the size constraints of the Shuttle payload bay. Thus, the architecture can utilize this heavy lift and large diameter to create fewer elements with more capability each. Current capabilities on the market utilize a ~7m fairing diameter with a total length of 18m to almost 22 m in length with larger fairings planned for future development. This, however, does include the conical section at the top of the fairing and thus total length will be less than these values. Early configurations of the station (e.g. the core module) will also be limited by ISS allowable external volumes due to keep out zones.

ISS was able to perform complex assembly tasks due to the utility of the Space Shuttle and its ability to support independent robotics and EVA operations. For initial operations at ISS, ISS transportation, robotics, crew, EVA and other services can be used for outfitting and checkout of the new platform. After separation, the assembly of the new platform must rely on internal connections for power, thermal, data, fluids and ventilation as EVA capability will not be available initially until an airlock is installed and activated (if desired in the architecture).

Robotic capability will be included to facilitate subsequent assembly once separated. The timing of when robotic capability would be included in the new station is a trade that requires future assessment. From an architecture standpoint, robotic capability allows for the use of berthing ports between modules. Berthing ports are considerably larger than docking ports and allow for transfer of large items, ventilation, coolant exchange and passing a large amount of power. Robotic capability would also enable the use of external research facility platforms and external spare ORU storage and retrieval.

There is also an open trade for including EVA in the architecture. The designs would include external provisions for EVA such as handrails and foot restraint attach points so as not to preclude a later decision. Depending on these trades, if no or limited capability exists externally, on-orbit replaceable units (ORUs) will be positioned internally, to the extent possible, to support maintenance and replacement. This is a

driver for overall habitable and accessible volume. Depending on the EVA trades, external versus internal ORU placement can be optimized in the architecture for anything that is beyond the expected robotic capability.

The architecture includes research plug-in locations in the core, commercial modules and future expansion modules, and provide power, active heat rejection, air cooling, vacuum and GN2 sources, video, and command and data handling for payloads and commercial activities. Any “permanent” facilities will be limited to those required to support continuous user needs. As a lesson learned from ISS experience, permanent facilities are inefficient as the locations can’t be flexibly used for emerging research demands and often have lower overall throughput rates. Input from industry will be considered in the layout and backbone resources included.

In the event ISS is not available for the core to attach to, it will become necessary to outfit the core module to sustain crew and basic research until later modules could be added. Much of the existing core design would still be utilized, but some systems such as life support, crew habitation, and robotics would need to be included in the initial launch and checked out on orbit. Further, more frequent crew and cargo flights, that include propellant resupply. Additional outfitting that was assumed to come from ISS would also need to be flown and installed (if not launched with) via these flights. Assembly of the commercial modules could still occur with the core but might be power/thermal constrained until the arrival of later elements that provide resources. In this case, however, it will be possible that the commercial modules generate their own power and reject heat to enable their commercial goals without the previous restrictions on configuration put in place by the ISS geometry.

1.3 USE OF ISS AS A PLATFORM:

There are significant benefits associated with leveraging ISS capabilities to establish a new space station in LEO. Leveraging ISS resources such as orbit altitude maintenance, robotics, EVA and IVA outfitting can mitigate early challenges with assembly and anomaly resolution, as seen with prior space stations. For example, ISS system anomalies that occurred during assembly were addressed by Shuttle crew and capabilities. Other early lessons learned during early ISS operations include the need for robust supply chain for spares and routine logistics/spares resupply, as well as having dissimilar systems to mitigate impacts of common mode failures. Assembly and checkout of the core module at ISS allows incremental build of capabilities, allowing for some deferral of select long lead systems, such as regenerative ECLSS and additional augmented power/thermal.

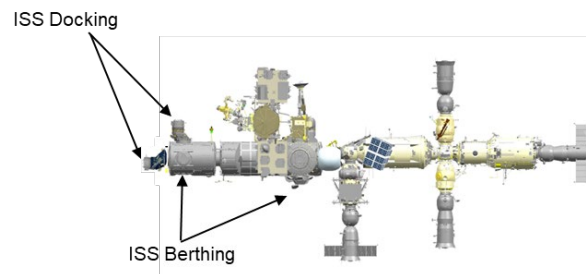


Figure 1: Current ISS Docking and Berthing Ports

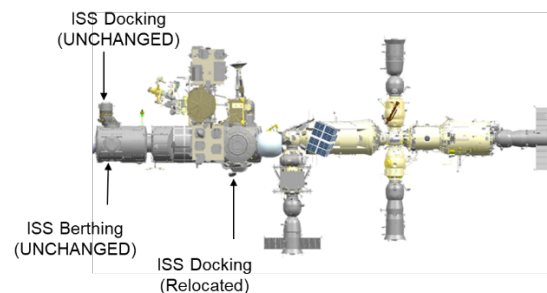


Figure 2: Proposed ISS Reconfiguration to allow for use of the Node 2 Forward Berthing Port

To allow for use of a core module, ISS must be reconfigured to allow the necessary services at the forward port while maintaining capability to return to its baseline configuration after separation of the new element to allow for safe ISS disposal utilizing the USDV. To leverage the ISS Node 2 Forward Port, a reconfiguration of ISS is required to move the Pressurized Mating Adapter currently at Node 2 Forward to the Node 1 Nadir port. Additionally, internal reconfiguration inside Node 2 and other ISS modules would be necessary to ensure that the appropriate utilities are routed to the Node 2 Forward Bulkhead, using a Common Berthing Mechanism that maximizes crew IVA connectors. Figure 1 and 2 shows ISS before and after reconfiguration. Various ISS resources could be made available at the Node 2 Forward Port. In addition to EVA, EVR, transportation and crew time, ISS provides other resources that could benefit the initial core module and the connecting modules prior to separation, including life support, commanding/monitoring, data downlink, communications, power, and cooling.

1.4 ARCHITECTURE DESCRIPTION

CORE MODULE

A NASA owned core module will be launched as the first element and will include minimum critical capabilities that can be evolved later. The core module will include propulsion, initial power capabilities, communications, command and data handling, navigation, avionics/software and heat rejection. The propulsion system will be designed to meet the full configuration needs of the future space station with the Core serving as the primary propulsion module for the on-orbit life along with the eventual deorbit. The Core will have scarred interfaces for future evolution of additional capabilities. The Core's power and heat rejection systems will be sufficient to provide for basic capabilities upon separation from ISS including basic life support systems and two attached commercial modules. During initial joint ISS operations, initial habitation life support inside the Core Module will be minimal leveraging that of ISS, but will include scarring to allow for later outfitting once separated. A conceptual external configuration of the Core Module is shown in Figure 3.

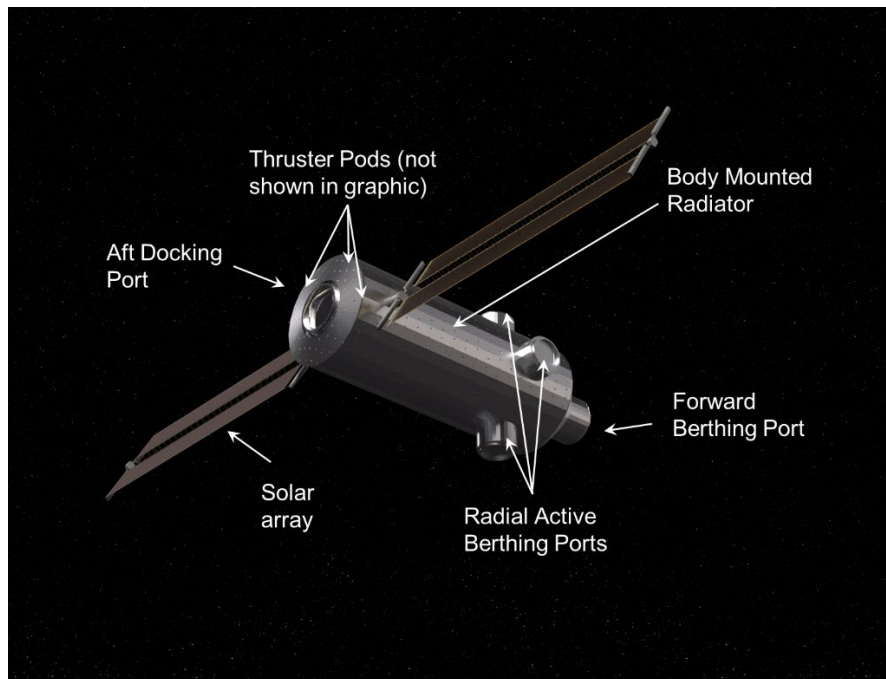


Figure 3 – Conceptual Core Module Layout

Figure 4 depicts a side view of ISS with the expected ISS port configuration with the Core Module berthed to ISS Node 2 Forward. NASA plans to maintain two docking ports (Node 2 Zenith and Node 1 Nadir) and one berthing port (Node 2 Nadir) to support ISS-funded visiting vehicle traffic. NASA

visiting vehicle traffic is not expected to use any potential berthing or docking ports on the Core Module, but this can be revisited as needed in the future.

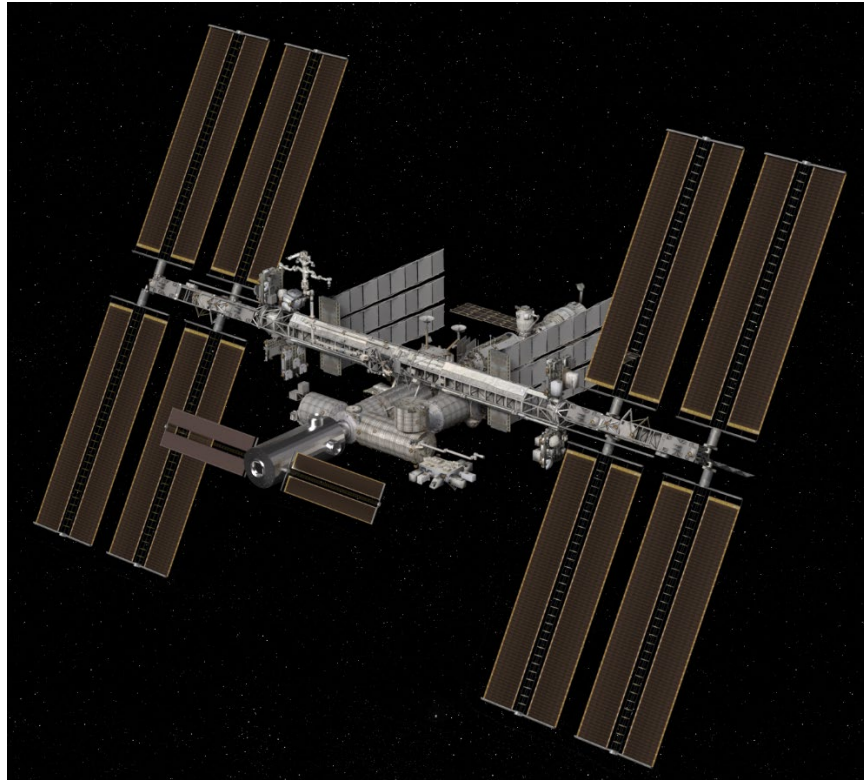


Figure 4 – Core Module Berthed to ISS

The Core's free flier propulsion capabilities will include stack attitude control, altitude maintenance, propellant storage, ability for propellant transfer from a cargo vehicle attached to the aft port and deorbit capability. Propulsion capability could also include supplemental electric propulsion to offset propellant demands, but this requires a trade against overall power availability and architecture complexity. The core module will also include control moment gyros to offset the propulsion demands for attitude control and maneuvers.

The core module's aft docking port will be used to transfer internal cargo, transfer propellant to the core module's storage tanks, and to provide propulsive control of the station from a cargo vehicle. The forward portion of the core module is an integrated Node with four radial ports and a forward port. One of the Node's ports will be used for a subsequent module that will provide expanded power generation and cooling, among other core functions such as regenerative life support. The radial ports will contain a combination of active CBM ports for compatible modules and docking ports for crew and cargo vehicles. Note that the forward port that is attached to ISS would require a passive common berthing mechanism whereas other ports will need active common berthing mechanisms.

The core will also have limited space for crew habitation and basic research. It is envisioned to have ability to connect to EXPRESS Locker Equivalents and potentially International Standard Payload Rack (ISPR) interfaces, dependent upon the needs from NASA and commercial interests. The extent of payload facilities will be dictated by need and constrained by resources and volume in the module, considering the other systems mentioned.

Key architecture trades for the core module include solar electric propulsion (SEP) to offset propellant demands, EVA and the timing of the when robotics is available on the new station. Additionally, the module size, and the configuration of attached modules (NASA and Commercial) and visiting vehicles would need to be considered against the existing constraints of ISS, such as the use of the nadir visiting vehicle capture envelope and related volume constraints, docking corridors, communication line of sight, keep out zones, etc.

Key capabilities for the Core Module include:

- Propulsive capabilities for free flier Orbit Maintenance
- Ability to be refueled from a visiting vehicle at the aft port
- Initial power and heat rejection
- Four radial and two axial ports for future module expansion and crew/cargo vehicles, with capability to berth two commercial modules.
- Launch with docking adapter(s) as needed to evolve station construction and crew/cargo traffic
- Support commercial module interfaces at module-based ports
- Size/mass to maximize use of large class launch vehicle fairings
- Arrives with internal propulsive capability
- Accommodates cooling fans, internal ventilation
- Modest secondary structure to support addition of ISS systems and Payloads or lockers

COMMERCIAL MODULES

The commercial modules would be a hub for research, commercial activities and additional crew habitation and life support functions, as determined by the commercial partner and NASA. The initial Core Module configuration, while berthed to ISS, can accommodate two commercial modules as shown in Figure 5. These modules will receive limited power, heat rejection and other resources from the core while either in free flight or at ISS. Resources will be constrained until supplemental power and cooling can be provided with future expansion.

The commercial modules would have the ability to use the radial common berthing mechanisms of the core module and the later power and cooling module. The forward port on the Core Module could also be made available depending upon operational constraints and NASA needs of the port. These modules would either transit to ISS using integrated propulsion or a commercial bus, depending upon the solution chosen by the commercial module provider, leveraging a robotic arm for capture and installation. At ISS, volume would be made available in the port and starboard directions as shown, with cutouts to avoid other ISS modules and interference with other ISS operations. The ability to separate these modules

from the core would be made available, as these modules were desired to transition to an additional free flying platform.

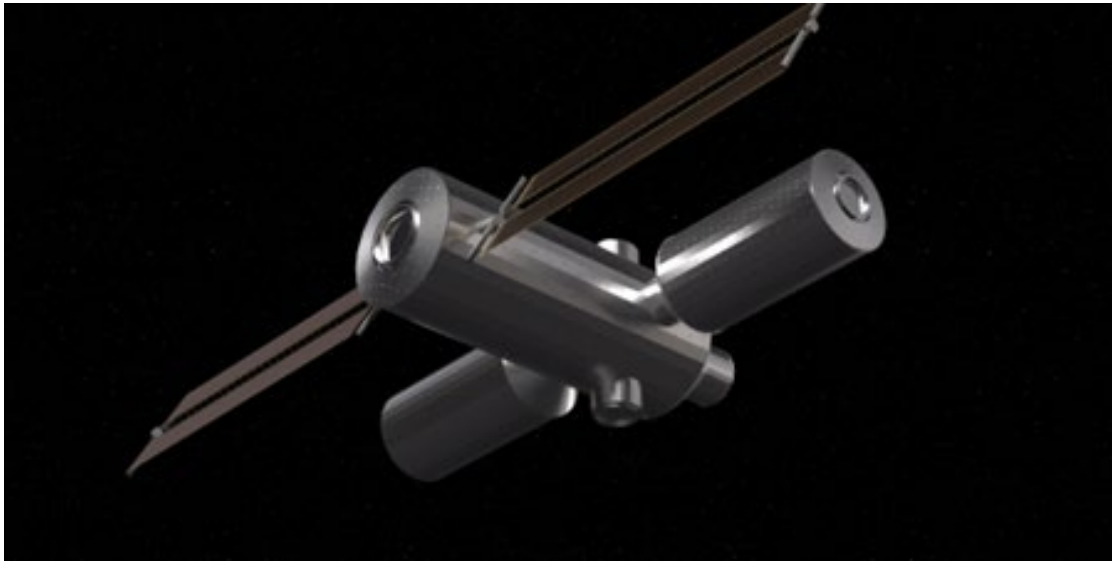


Figure 5 – Core Module with Additional Commercial Modules (shown in green). The remaining ports are reserved for visiting vehicle traffic and future expansion.

POWER AND COOLING MODULE (PCM)

An additional module with a high-capacity heat rejection system and a power system with equal or larger arrays and additional battery storage capability will be required to support the full utilization of the station’s capabilities as well as continuous human presence. Considerations for overall power demand will include NASA’s utilization needs in addition to any international and commercial needs. This Power and Cooling Module (PCM) (shown in Figures 6 and 7) could either be installed on the zenith port of the Core Module with arrays that have an Alpha and a Beta articulating joint or on the forward end of the Core Module with single axis articulating arrays. This module is intended to provide additional resources and volume to aid commercial development and to satisfy NASA needs for continuous crew, research and commercial capabilities. Launch vehicle fairing capability will dictate acceptable sizing for this module. The PCM may utilize the Core design to minimize new development or a new design to optimize station layout, power generation and heat rejection.

The internal functions of the PCM will include expanded habitation capabilities, regenerative ECLS and could include some research and manufacturing backbone capabilities for interoperable experiment “plug-in” locations. Priority for research and manufacturing capabilities will be given to the commercial modules and any specific content would factor in any commercial module capabilities planned for the station. If a robotic arm external payloads are desired, the PCM may also include exterior attach points or a pallet for payloads. If market demands have a need to provide research capabilities, then commercial arrangements for facility use, sharing or capabilities would be offered.

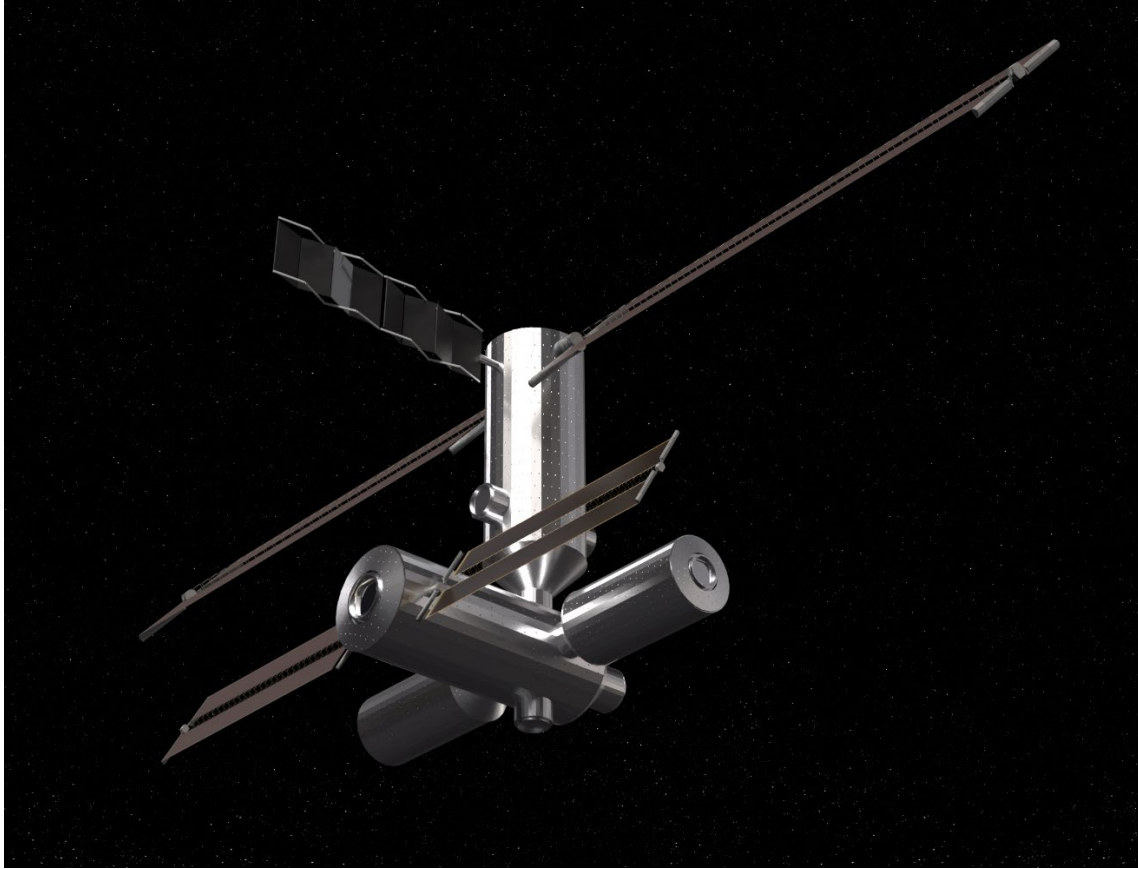


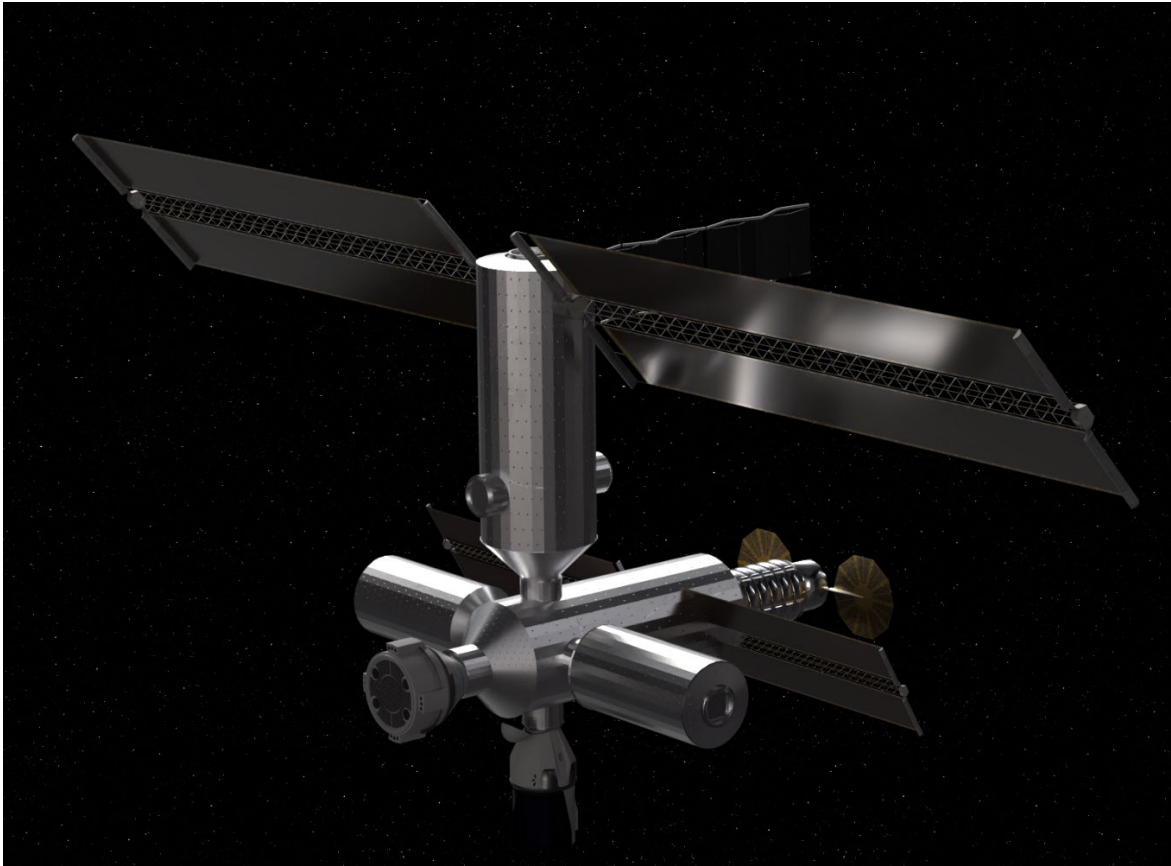
Figure 6 – Concept PCM on Core Zenith Port

ROBOTIC ARM

The new Space Station will need a robotic arm that is capable of performing track, capture and berthing of modules, module relocations as needed, transfer of external payloads and external ORU R&Rs. Modules would need to be outfitted with appropriate grapple fixtures to enable robotic handling and sufficient external video camera views would also be needed. The Core Module and commercial modules attached while on ISS would require an ISS SSRMS compatible grapple fixture. External ORU's slated for robotic maintenance would need standardized robotically compatible interfaces, alignment guides and cues, etc.

ADDITIONAL MODULES

Additional commercial and other modules could be added to the radial ports of the PCM. Note that these modules may have power/thermal and volumetric constraints given the configuration of the core and power/thermal module and their relative performance. An example is shown in Figure 8 below. The layout and capabilities provided to these modules, along with the inherent functions and capabilities of the modules will be guided by commercial industry input and NASA needs in LEO.



1.5 KEY ARCHITECTURE TRADES

There are a number of architecture trades that have been considered in the concept for a new station. Rationale for trades that have been concluded along with open trade areas are summarized below.

BERTHING VERSUS DOCKING

Because the Core has a berthing port for compatibility with the ISS, a robotic arm is included in the architecture and module-to-module attachment is conducted via berthing. As a goal, NASA would like to see non-crewed, automated track and capture of arriving modules. This could be through a combination of ground control along with navigation and software aids for the robotic arm. A berthing port allows for a much larger interior pass-through along with commodity sharing between modules without the need for drag throughs or EVA connections. These commodities are passed through the vestibule and can include ventilation, coolant exchange, power, data, video, and audio.

For crew and cargo vehicles, docking is the preferred attachment method. There are currently vehicles designed as berthing vehicles that would need to be converted to docking vehicles. If all vehicles utilize docking systems, the ports are more extensible freeing up ports and openings in the flight plan. The downside of docking is the cost and the active unit flying on a disposable cargo vehicle which increases the overall mission cost. This is still an open trade area.

It should also be noted that current docking system standards and designs do not accommodate module to module commodity exchange which presents challenges for the architecture, with either extensive drag-throughs, external connects (EVA driver), or duplicated equipment to provide the functions (e.g. each module with cooling, CO2 scrubbing, etc). This is a key driver for using berthing between modules.

Docking and berthing ports will need to include adequate clearances to both the early and final solar arrays and radiators to allow for mass optimization of those structures. Plume loads from visiting vehicles create significant loads on exposed appendages so the design needs to accommodate these loads or considerable docking and berthing constraints will exist and severely limit the flexibility in vehicle flight planning.

NASA may elect to maintain flexibility to trade berthing versus docking adapters until later in the life cycle, when the cost/schedule/technical trade results come into focus, which may necessitate the development of a generic interface on the primary structure to allow for flex as the design matures.

ROBOTIC ARM

The new robotic arm would need capabilities to capture free flying modules and visiting vehicles and install them, along with the ability to help remove and replace external ORUs and walk off to other modules to continue the assembly process. When the new arm arrives in the architecture is flexible. Ideally the robotic arm would launch with the Core Module along with the video, avionics, and control needed to operate the arm. It is technically feasible to launch the robotic arm with the first free flying module, but this would be a riskier mission configuration without early checkout of the arm's functionality before it is needed for its first track and capture. The track and capture on ISS would utilize the existing ISS robotic arm.

EVA

EVA capability would add capability to mount ORUs externally allowing for replacement, service/repair of external systems (such as robotics), and upgrades to external systems. If EVA capability is included in the architecture, an airlock with an equipment lock, crew lock and a docking or berthing port would be included. This airlock would consume one of the available ports. The location and delivery timeline of an airlock would need to be traded.

SOLAR ELECTRIC PROPULSION (SEP)

Propulsion capability could include solar electric propulsion (SEP) to offset propellant demands, but this requires a trade against overall power availability and architecture complexity. In-space electric propulsion systems, such as SEP, have extremely high fuel economy at lower thrust, providing mission flexibility and capabilities not achievable using traditional propulsion systems.

SEP systems use approximately 10 times less propellant than a comparable conventional chemical propulsion system, such as those used to propel the Orion spacecraft or space shuttle. The reduced fuel mass significantly reduces launch costs while delivering robust in-space propulsion capable of sending robotic and crewed missions well beyond low Earth orbit to distant destinations, like the Moon, Mars, and the outer solar system. Significant power is required and other affects such as self-pluming, percentage

of the orbit when SEP could be used given battery recharge needs, thrust compared to drag of the station, propellant resupply would need to be traded.

CORE MODULE SIZE

The core module size (diameter and length) should take advantage of advances of launch capability in the commercial launch market and to limit number of launch and on-orbit assembly steps to reduce cost and complexity. Interior volume for vehicle systems and research will be valuable, in addition to the inclusion of some ORU boxes to minimize need for external robotics or EVAs (this is a future trade). Depending upon overall station configuration and needs, constraints will be developed as to module size and installation location. At ISS, the core module and addition of attached modules to the core would also need to be considered against the existing constraints of ISS, such as the use of the nadir visiting vehicle capture envelope and related volume constraints, docking corridors, communication line of sight, keep out zones, etc.

1.6 ASSEMBLY SEQUENCE AND INITIAL OPERATIONS:

1.6.1 Operations at ISS

The ISS will be reconfigured as described above in advance of the Core Module's launch. The first launch will deliver the Core Module to Low Earth Orbit near the ISS's operating altitude range. Shortly after separation from the launch vehicle the core module will automatically deploy its solar arrays as well as communication antennas. Following checkout of critical systems, it will raise its orbit to the operating altitude of ISS. The Core Module will utilize its own propulsion and navigation capability to rendezvous with the ISS. Upon performing successful proximity operations and reaching the ISS capture volume, the Core Module would be captured by the Space Station Remote Manipulator System (SSRMS) and subsequently berthed at the Node 2 Forward location using the common berthing mechanism. Once the vestibule between ISS and the Core Module is pressurized and leak checked, the hatches of both the ISS and the Core will be opened and the ISS crew can ingress.

The ISS crew will mate the initial vestibule connections to provide the Core with critical keep alive power. A subsequent ingress of the Core will be conducted after MCC confirms that all systems are performing nominally. Subsequent crew activities include establishing power generation, confirming power management and distribution, communication with ISS and the ground, cooling to racks and other system function checks will be performed. The Robotic Arm, if launched with the Core Module and available, will have its launch locks removed by either remote commanding or through action by the ISS crew and tested out.

Though the Core's solar arrays will be deployed and mandatory Core systems will be checked out in free flight prior to ISS berthing. ISS will supplement power to the Core while it is attached. The arrays will need to balance the core modules future power needs upon separation from ISS, supporting attached modules, and adhering to ISS volumetric and other operational constraints.

Gradual outfitting of the core module would follow through ISS cargo resupply and utilizing ISS crew. It is anticipated that some equipment may be launched in a soft stowed configuration in order to limit the

launch loads on equipment and the complexities associated with building structure to support equipment in the launch environment. ISS cargo deliveries, both prior to the Core's arrival and after its arrival will be used to deliver additional outfitting equipment as needed.

Commercial modules will attach to the Core module at ISS. They will be launched with power, propulsion, navigation and communication capabilities necessary to safely rendezvous with the ISS. The ISS robotic arm will be used to capture and berth the modules to radial ports on the Core Module. The core module forward port could also be used within operational constraints and NASA needs of the use of that port. Once attached, the module can be outfitted with any ISS hardware and facilities that are planned for transfer to the new station. Power and cooling capability will be limited by what ISS can provide to the Core and commercial module, so powered loads will need to remain low, unless the commercial modules can supplement these resources.

During joint operations, ISS ground control teams would maintain overall command, monitoring and control functions of the core module. The extent of the control and monitoring of the commercial modules will be defined based on functions, necessity for ISS crew and vehicle safety, and commercial mission control capability. Current or evolved communications systems would be used to downlink telemetry and data. Existing ISS control infrastructure will be maintained, however commercial providers may choose to have additional control centers that monitor commercial module health and other activities, and these control centers will work with the ISS mission control team.

1.6.2 ISS Separation and ISS Reconfiguration

After the Core Module is outfitted and commercial modules are added, the integrated performance of the stack along with key functionality will be confirmed. New station reconfiguration may occur to ensure sufficient and viable docking and berthing ports. This might require a specialized docking adapter to be launched and staged to either attach to the passive CBM or to an active CBM interface on the core modules forward port (in the latter case a passive to passive berthing adapter would need to be flown on the initial core module as the core and the ISS both have active CBM interfaces). Once confidence is gained that the new core can function on its own and the requisite outfitting has been performed, the Core with connecting modules will be unberthed from ISS at any time to become a free flyer.

Following the new station's departure, the ISS forward port would be reverted back to the current configuration to allow for docking, ISS altitude would be lowered and the US Deorbit Vehicle (USDV) would be launched when ISS reaches its target altitude followed eventually by the final deorbit maneuver.

1.6.3 New Space Station Free Flying Operations

Once separated, independent operations of the new station will be re-verified, including testing of critical systems such as propulsion, attitude control and power/heat rejection in free flight. The vehicle will continue to be controlled from NASA mission control with support from the commercial partner ground control assets. At this point, it is anticipated that sortie missions will begin to the independent space station. A sortie mission will involve first the delivery of a cargo vehicle that will remain attached for the duration of the sortie. This cargo vehicle will provide the necessary supplies including life support

consumables, water, food, waste disposal containers, and any science hardware planned for the mission. Once this vehicle is docked the crewed mission can launch and dock to the station to conduct the prescribed mission. This mission can include government objectives, commercial objectives or both depending on the market needs and duration will be driven by customer needs. Missions will continue regularly in this fashion, and we anticipate at least two per year with more possible depending on demand and funding. Opportunities to cost share missions and allow for increased frequency of sorties will be possible with outside commercial investment and NASA will look for input from the overall industry input to inform the government on how best to proceed.

If NASA or Commercial needs drive the delivery of additional power and cooling capability, continuous long duration human missions, or new external attach locations, a supplemental power and cooling module would be launched to the station. This module would likely be captured by the station's robotic arm and berthed to the appropriate port. This new module could either deliver or enable the delivery of regenerative ECLSS systems that would allow for continuous human presence onboard with significantly reduced resupply needs for water, waste disposal containers

Through the lifetime of the new space station, the propulsion systems on the core would maintain station orbit maintenance and conjunction avoidance, supplemented by docking vehicles that would provide propellant resupply, attitude control and additional translational control to extend the life of station systems. The crew would use onboard EVR and EVA resources (if included), to maintain, upgrade and expand the station. To gain the ability to transfer crew and cargo externally, an airlock would need to be included. Operations of the airlock would be similar to that of the ISS today. Crew and cargo resupply would continue to bring new crews, research, spare parts, consumables and other logistics. This will be discussed further in the next section.

Commercial modules may continue to use this platform and may choose to eventually separate and become their own free flying space station. This operation would be similar to that of the core module separating from ISS, with exception of the new robotic arm, or an undocking capability being used to facilitate the separation.

At the end of life, this new space station would use the propulsion capability to deorbit safely back to Earth.

1.7 TRANSPORTATION

Transportation will be based on the current designs and interfaces for docking and berthing vehicles. It is expected that the capabilities of the vehicles will be similar to what exists for ISS today with the addition of propellant delivery/transfer and reboost from the Core module's aft port. The station will accommodate at least 3 ports for cargo and crew vehicles. Crew and cargo vehicles will need modifications for new communication systems for space to ground as the TDRS network is phased out. These systems do not necessarily need to be compatible with the station's space to ground communication system. It is expected that the space-to-space communication system will maintain the same specifications as today's C2V2 system.

In addition, cargo vehicles will need modifications to transport and transfer propellant, and to provide altitude (translation) control of the stack once docked. Flexibility will be needed to trade propellant and cargo loading, in some cases NASA may need to maximize propellant and trade on upmass.

Transportation Cadence - Crew

Crew transportation is envisioned to be on a 6 month cadence in line with ISS experience today through commercial transportation providers. This will involve a crew vehicle always attached for the 6 month duration. For crew rotation, the new crew will arrive on their vehicle prior to the departure of the previous vehicle and crew. This direct handover period is anticipated to be on the order of a week for nominal cases however landing weather has been shown to impact this duration. While handover periods are not mandatory, experience has demonstrated an increase in early productivity of the crew with effective handover on station systems and configuration. Handover periods will be adjusted situationally based on overall operations and resource needs. Depending on available resources either through government funding or via private commercial means, additional short duration missions will be possible.

Transportation Cadence - Cargo

Cargo transportation will also occur through commercial services similar to that of the ISS Commercial Resupply Services contract today with the opportunity for international partners to contribute as desired. To effectively maintain the crew, vehicle and science resupply needs a regular cadence of cargo flights will occur throughout the year. ISS operated with 4-5 cargo vehicles per year on the US side to supply the needs of 4 crew onboard. These flights will consist of a balance of smaller return capable vehicles and larger disposal vehicles that burn up upon reentry. The return vehicles are anticipated to stay for shorter durations allowing for sortie payload opportunities and regular return of failed components from the station for refurbishment. Typical duration for these missions will be 30-45 days but could vary based on specific scientific or commercial needs. Larger disposal vehicles will stay longer typically for ~ six months. During this time, they will serve as a staging area for trash disposal allowing for more space to be maintained inside the station for nominal items. Lessons from ISS operations inform a paced approach for cargo offload and trash loading for vehicles of this type due to their larger capacity. Propellant resupply will be an added need that does not exist within the current fleet of ISS transportation vehicles. The approach utilizes aft cargo vehicles for propellant delivery as well as supplemental propulsive attitude and altitude management. Utilizing disposable cargo vehicles in this capacity allows for the core system propulsion capabilities to be saved and life extended. Propellant delivery needs will increase with solar activity and thus will drive the cadence of vehicles to the aft port during peak years of solar activity. During peak years it is anticipated that additional flights will be needed to support propellant needs of the station thus driving additional flights. It is also expected that these vehicles will transfer any remaining prop to the Core Module prior to departure.

1.8 COMMERCIAL EXPANSION

A core function of this space station is to stimulate commercial industry and set the stage for multiple free flying space stations in LEO. As mentioned previously, NASA will make available several expansion opportunities that build off the core and power and cooling modules. Up to two can be attached to the Core Module at ISS. Post ISS separation and with the addition of the Power and Cooling Module, multiple additional commercial elements can be attached. The space station stack can ensure survival until activation and checkout can be performed allowing commercial companies to assemble their base capabilities here before departing. The architecture is designed with future extensibility in mind creating multiple opportunities for commercial elements of different shapes, sizes, and capabilities. Each element

will receive some base level of keep alive power, but each module will need to provide supplemental power and heat rejection.

Following the arrival of the PCM, expansion ports will be available for entities while maintaining at least ports reserved for transportation vehicles. Additional expansion is also possible, building off the forward Core Module. Use of these ports will be industry driven and could comprise of individual or partnered approaches, yielding 3+ modules attached. Alternatively, a commercial segment with one or many commercial partners could also be supported. Multiple partnering opportunities will be made available.

Depending on available resources either through government funding or via private commercial means, additional short duration missions will be possible.

Future commercial expansion opportunities and associated enabling capabilities and timing will be informed by industry and NASA needs. NASA will continuously partner with industry to identify the best ways to ignite a commercial market.